Multi-modal Multi-agent Debate-based Dialectical Knowledge Transfer Learning

Anonymous ACL submission

Abstract

Transferring knowledge from large to smaller models can enhance their reasoning abilities and expedite deployment. However, for multi-modal complex reasoning tasks, existing large models often produce knowledge with factual errors. Transferring this flawed multi-modal knowledge to smaller models can lead to error accumulation, thereby degrading their performance. To address this challenge, we propose a novel Multi-modal Multi-agent Debate-based Dialectical Knowledge Transfer Learning framework (**M2D-DKTL**). Initially, we introduce a multi-modal multi-agent debate framework to uncover deep logical relationships and enhance knowledge quality. Then, we propose using chain-of-debate to condense dialectical knowledge and transfer it to smaller models, thereby improving their complex reasoning ability. Experimental results show that debates can enhance the model's understanding and reasoning in complex scenarios. Moreover, our method achieved average improvements of +4.3%, +3.8%, and +2.6% on the MMMU, MathVista, and CMMMU datasets, respectively, validating its effectiveness and generalizability. Further analysis reveals that our method enhances the logical reasoning and deep analysis capabilities of smaller models.[1]

1 Introduction

Multi-modal large language models (MLLMs) have demonstrated remarkable potential across various multi-modal tasks (OpenAI, 2023a; Team et al., 2023), such as reasoning (Liu et al., 2023b; Hong et al., 2023), retrieval (Wang et al., 2023b), and captioning (Li et al., 2023a; Wang et al., 2022a). Smaller MLLMs have drawn significant attention within the research community due to their lower deployment costs and faster inference speeds (Liu et al., 2023b; He et al., 2024). However, when the question surpasses the knowledge boundaries of MLLMs, these models can produce factual errors, known as hallucination (Huang et al., 2023), which severely impact their reliability and usability in practical applications.

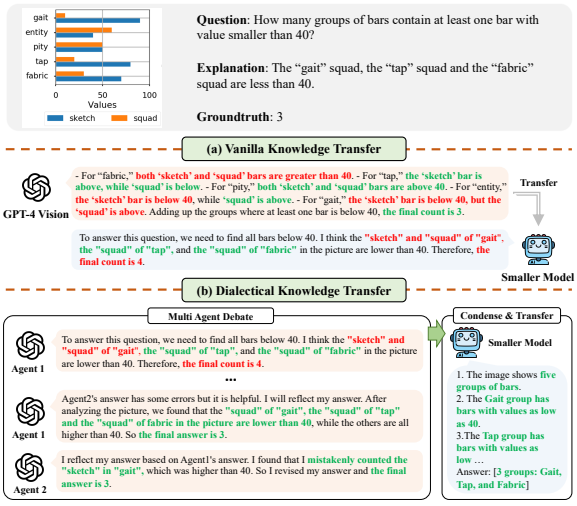


Figure 1: **Red bold** indicate incorrect reasoning, and **green bold** indicate correct reasoning. Knowledge with factual errors can lead to error accumulation (a). Multi-agent debates can generate high-quality dialectical knowledge, effectively enhancing the reasoning abilities of smaller models (b).

Current methods transfer knowledge from large to smaller models to improve the performance of the latter (Fu et al., 2023b; Ho et al., 2023; Li et al., 2022). They prompt a large model to generate rationales for downstream datasets, which is then used to train smaller models (Hsieh et al., 2023; Wang et al., 2023c). However, multi-modal complex reasoning tasks require a comprehensive understanding of multi-modal information and accurate reasoning (Yue et al., 2023; Lu et al., 2023). Even advanced models such as GPT-4 Vision (OpenAI, 2023b) frequently suffer from hallucinations, producing reasoning processes with factual errors. As shown in Figure 1(a), transferring knowledge with factual errors to smaller models can lead to

[1]We promise to release all data and code after publication.

error accumulation, thereby degrading their performance. Therefore, reducing factual errors and improving knowledge quality are critical challenges for knowledge transfer learning.

Recently, multi-agent debates (MADs) have shown outstanding performance in complex reasoning tasks (Fu et al., 2023a; Qian et al., 2023; Chan et al., 2024). They organizes multiple agents to challenge and revise each other's viewpoints, uncovering deep logical relationships and enhancing knowledge quality (Du et al., 2023; Liang et al., 2023). As illustrated in Figure 1(b), multi-agent debates can reduce factual errors in the reasoning process, condensing them into dialectical knowledge and transferring them to smaller models to address the aforementioned challenges.

Inspired by these motivations, we propose a novel multi-modal multi-agent debate-based dialectical knowledge transfer learning framework (**M2D-DKTL**). First, we introduce a multi-modal multi-agent debate framework (M2D) that uncovers deep logical relationships between different modalities through debates, thereby enhancing the quality of knowledge. Then, we propose using a chain-of-debate to condense and optimize dialectical knowledge into a format suitable for fine-tuning. Finally, we transfer this knowledge to smaller MLLMs using supervised fine-tuning methods, thereby enhancing their complex reasoning capabilities and accelerating their practical application.

Experimental results show that our debate framework outperforms the best baseline methods on MMMU (Yue et al., 2023) and MathVista (Lu et al., 2023) by +6.4% and +3.2%, respectively. This demonstrates that debates can uncover deep logical relationships between different modalities, enhancing the model's understanding and reasoning capabilities. Subsequently, our knowledge transfer framework achieved average improvements of +4.3%, +3.8%, and +2.6% on MMMU, MathVista, and CMMMU (Zhang et al., 2024), respectively, validating the effectiveness and generalizability of our method. Further analysis indicates that our method enhances the smaller MLLMs' logical reasoning and deep analysis capabilities.

The contributions of this work are as follows:

- We propose a multi-modal multi-agent debate framework that can uncover deep logical relationships between different modalities, thereby improving the quality of knowledge.
- We condense the dialectical knowledge into chain-of-debate and utilize transfer learning to endow it to smaller MLLMs, thereby enhancing their dialectical reasoning abilities.
- Experimental results demonstrate that our method improves the logical reasoning and deep analysis capabilities of smaller models, validating the effectiveness and generalizability of our approach.

2 Related Work

2.1 Multi-Modal Agent

The introduction of advanced MLLMs like GPT-4 Vison (OpenAI, 2023a) and Gemini (Team et al., 2023) has marked a significant shift in the field, garnering widespread attention. These models are now increasingly used for visual understanding tasks (Sun et al., 2023; Bai et al., 2023; Hong et al., 2023). Liu et al. (2023a) trained the LLaVA with data constructed based on GPT4, showing impressive visual reasoning performance. Further, several researchers have explored their application in tool usage for more complex tasks (Gao et al., 2023; Liu et al., 2023c). Liu et al. (2023b) developed a multi-modal tool-calling agent based on LLAVA. In contrast, our method leverages high-quality dialectical knowledge generated through MADs further to enhance the dialectical reasoning capabilities of smaller MLLMs, thereby accelerating their real-world application.

2.2 Multi-Agent Debate

The emergent capabilities of LLMs (Wei et al., 2022a) have demonstrated remarkable in numerous natural language processing tasks (Pu et al., 2023; Touvron et al., 2023). To mitigate the hallucination challenge, recent research has focused on constructing multiple agents based on LLMs to facilitate complex reasoning through debate (Fu et al., 2023a; Xiong et al., 2023; Wang et al., 2023a). Du et al. (2023) and Liang et al. (2023) innovatively enhanced model consistency via multi-agent debates, substantially boosting performance in complex reasoning tasks. Chan et al. (2024) developed a multi-agent debate-based approach for evaluating natural language generation tasks. The dialectical knowledge generated by existing multi-agent debate methods cannot be directly applied to knowledge transfer learning. Thus, we propose condensing the debate content produced by various agents into a chain-of-debate, which would be appropriate for fine-tuning smaller MLLMs.

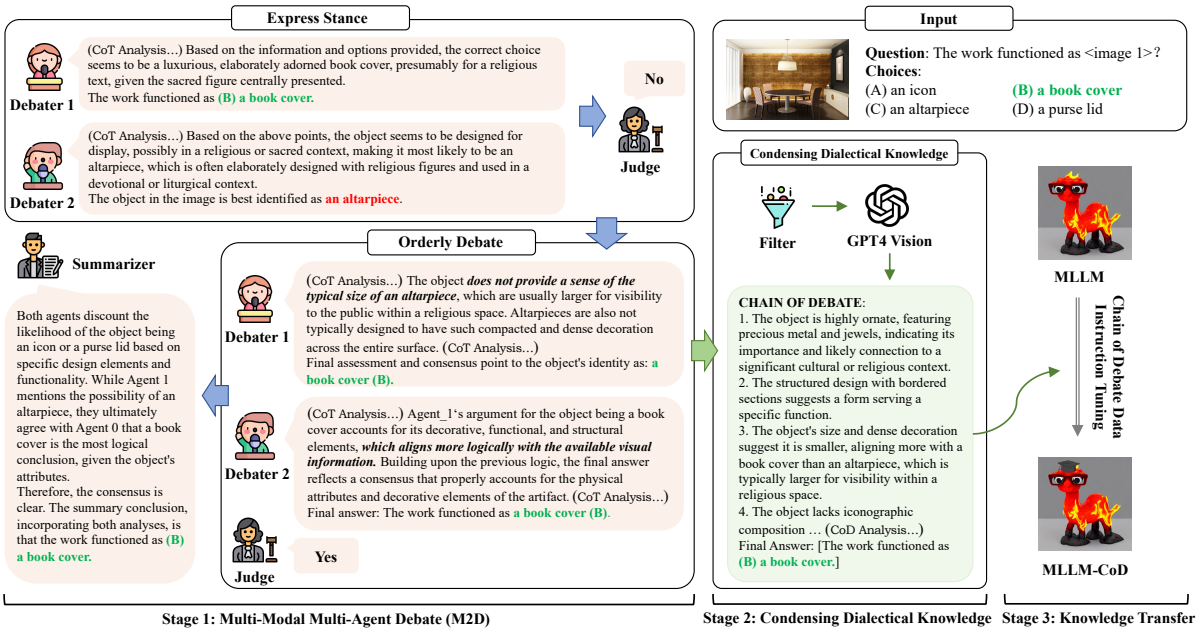


Figure 2: The framework of our dialectical knowledge transfer learning mechanism based on multi-modal multi-agent debate. **Stage 1**: Multi-modal multi-agent debate to enhance the quality of knowledge. **Stage 2**: Condensing the dialectical knowledge inherent in the debate process into chain-of-debate. **Stage 3**: Using knowledge transfer learning to endow dialectical knowledge to smaller MLLMs.

2.3 Knowledge Transfer Learning

Some recent works have explored knowledge transfer from large models to small models (Li et al., 2022; Wang et al., 2023c; Fu et al., 2023b). West et al. (2022) explored extracting structured knowledge from larger models to enhance small models, achieving improved commonsense reasoning. Inspired by the effectiveness of LLMs with CoT prompts, Li et al. (2023b) and Hsieh et al. (2023) employed CoT distillation from LLMs to enhance smaller models' reasoning abilities. In contrast to simply transferring knowledge generated by LLMs, we transfer high-quality dialectical knowledge produced by MADs to smaller MLLMs, enhancing their complex reasoning capabilities.

3 Method

3.1 Overview

As shown in Figure 2, our model framework comprises three stages. We propose a multi-modal multi-agent debate framework to improve the quality of knowledge in the first stage (§ 3.2), and the debate process is depicted in Algorithm 1. In the second stage (§ 3.3), we condense the dialectical knowledge inherent in the debate process into a chain-of-debate. Finally, in the third stage (§ 3.4), we use knowledge transfer learning to endow dialectical knowledge to smaller MLLMs. Note that when using MLLMs for reasoning, we did not distinguish between multi-choice and open questions in prompts but unified them into a generative reasoning question-and-answer format. The details of prompts and role definitions are shown in the Appendix A and G.

3.2 Multi-Modal Multi-Agent Debate

In this section, we define dialectical knowledge and the roles of agents and outline the debate procedure. The process is depicted in Algorithm 1.

3.2.1 Dialectical Knowledge Definition

The debate framework involves multiple agents in a structured debate, each presenting and defending their stance on a given question. Through challenging and revising these stances, agents collaboratively reach a resolution with a deeper understanding of the question, similar to human dialectical reasoning that synthesizes different viewpoints for profound insights (Slonim et al., 2021). Thus, we define "**dialectical knowledge**" as *the insights gained from debate and discussion among multiple agents*, characterized by fewer factual errors.

3.2.2 Role Definition

Our debate process contains three key roles: debater, judge, and summarizer. Their responsibilities are described below:

Debater: Debaters express their views on the questions and engage in debate to obtain correct and consistent answers.

Judge: The judge ensures the consistency of debaters' views, concluding the debate if consensus is reached; otherwise, it continues, maintaining efficiency and effectiveness.

Summarizer: The summarizer reviews the debaters' conclusions and forms an objective final summary without incorporating personal opinions.

3.2.3 Debate Process

For given a question q, we define the role set $\mathcal{R} = \{\mathcal{D}, J, S\}$ based on the MLLMs, where J represents the judge, S represents the summarizer, $\mathcal{D} = \{D_1, D_2, ..., D_m\}$ represents the debater set, m is the number of debaters. The debate process consists of two phases: *express stance* and *orderly debate*. During the express stance phase, each debater presents their stance independently. In the orderly debate phase, each debater updates their stances based on the debate history of the previous rounds and the question. The formula for expressing a stance is as follows:

$$s_i^r = \begin{cases} D_i(q) & r = 1 \\ D_i(q, s_1^{(r-1)}, ..., s_m^{(r-1)}) & r > 1 \end{cases} \quad (1)$$

where s_i^r represents the stance of debater D_i in round r.

Upon completing a debate round, all debaters $\mathcal{D}$ present their views, and the judge J determines if a consensus is reached. If so, the debate concludes, summarizer S finalizes the stance. Otherwise, the debate progresses to another round. As depicted in the *orderly debate* section of Figure 2, interactions such as $Debater1$ correcting $Debater2$ help refine and evolve the stances, fostering accurate dialectical knowledge.

3.3 Condensing Dialectical Knowledge

Agents correct each other's errors through debate, creating a dialectical logical structure that represents richer and more accurate knowledge. This structure is condensed into a CoD format for fine-tuning smaller MLLMs. This method enables smaller models to engage in comprehensive, dialectical thought processes, thereby enhancing their complex reasoning abilities.

3.3.1 Filter Nosiy Data

Since the transfer data did not include the true answers, we filtered out clearly incorrect data to avoid introducing noise. To minimize the effect of failed debate data on future transfer learning, we implement three filtering rules:

No Answer: Debates that did not reach a final result, such as those with reasoning results of None or No Answer, need to be filtered out.

Wrong Answer: In multiple-choice questions with a single correct answer, reasoning results presenting multiple options must be filtered out.

Closest Answer: For multiple-choice questions, if debate errors result in a non-option answer or the closest answer is chosen and needs to be filtered out.

Algorithm 1 Multi-Modal Multi-Agent Debate

Require: Complex reasoning questions, Q
Require: Multiple debaters based on MLLMs, $\mathcal{D} = \{D_1, ..., D_m\}$
Require: Judge based on MLLMs, J
Require: Summarizer based on MLLMs, S

```
for r in range(MaxDebateRound) do
    if r is 1 then
        for D_i in D do
            s_i^r = D_i(Q)
        end for
    else
        for D_i in D do
            s_i^r = D_i(Q, s_1^(r-1), ..., s_m^(r-1))
        end for
    end if
    if J(s_1^r, ..., s_m^r) is True then break
    end if
end for
ReasoningResult = S(s_1^r, ..., s_m^r)
return ReasoningResult
```

3.3.2 Chain-of-Debate

In multi-modal multi-agent debate framework, the debate process elevates complex reasoning by engaging multiple agents in an interactive process where they correct each other's reasoning errors, enhancing the overall accuracy and consistency of the reasoning results. This process creates a rich dialectical knowledge, which we condense into a CoD. Unlike the linear progression of the CoT (Wei et al., 2022b) approach, the CoD dynamically evolves through the debate, ensuring that the final reasoning reflects a consensus among all participating agents. This interactive error correction mechanism, unique to CoD, significantly reduces the likelihood of persisting errors that might oc-

cur in the CoT's solitary reasoning path, thereby providing a more robust solution to complex multi-modal reasoning tasks.

3.4 Dialectical Knowledge Transfer Learning

After obtaining training data containing dialectical knowledge, we employ the knowledge transfer learning method to teach smaller MLLMs the dialectical knowledge from multi-agent debates, endowing the small MLLMs with the capability for logical reasoning. For a given a dataset $\mathcal{X} = \{(img_i, q_i, y_i)\}^N$ with N training instances, where img_i represents an image, q_i represents a question about image, and y_i represents its answer. Similar to Wang et al. (2023c), in constructing dialectical data, we generate the dialectical reasoning process $\hat{d}_i$ and predicted answer $\hat{y}_i$ using content from multi-modal multi-agent debates. Then, during the knowledge transfer learning process, we input images img and question q to maximize the likelihood of the multi-modal small model to generate $(\hat{d}, \hat{y})$ as follows:

$$\mathbb{E}_{(img,q,\hat{d},\hat{y})} \log P(\hat{d}, \hat{y} | img, q; \theta) \quad (2)$$

where θ is parameter of the MLLMs.

4 Experiment

In this section, we conduct multiple experiments to show that multi-modal multi-agent debates can enhance performance on multi-modal complex reasoning tasks and generate high-quality dialectical knowledge (§ 4.1). Further experimental results demonstrate that incorporating dialectical knowledge can improve the logical reasoning capabilities of multi-modal small MLLMs (§ 4.2).

4.1 Multi-Modal Multi-Agent Debate

4.1.1 Benchmark

We selected two multi-modal complex reasoning datasets, MMMU (Yue et al., 2023) and MathVista (Lu et al., 2023), to evaluate the effectiveness of the multi-modal multi-agent debate method. MMMU covers six core disciplines, focusing on advanced perception and domain-specific reasoning, challenging models with expert-level tasks. MathVista combines diverse mathematical and visual tasks, requiring detailed visual understanding and compositional reasoning. Due to time and funding constraints, we sampled about 50% of items from each subcategory of the MMMU validation and MathVista testmini datasets to form the evaluation dataset. See details in Table 1.

Name	Language	Number	From
Inference Test			
MMMU	en	450	Validation Dataset
MathVista	en/zh	500	Testmini Dataset
CMMMU	zh	448	Validation Dataset
SFT Knowledge Transfer Learning			
MMMU	en	2,275	Test Dataset

Table 1: Datasets statistical results. Inference test dataset for evaluate model, SFT dataset for knowledge transfer learning.

To better evaluate the generation task, we referred to the evaluation methods (Zheng et al., 2023; Wang et al., 2023a), input the question, reference answer, and MLLM predicted answer into GPT-4, and let GPT-4 determine whether it is correct. Finally, we use accuracy to evaluate the results. See the Appendix G.2 for evaluation prompts.

4.1.2 Baselines

We selected the current best-performing MLLMs as our baselines:

Qwen-VL (Bai et al., 2023) is a multi-modal language model trained in three stages: pretraining, multi-task pretraining, and supervised fine-tuning.

CogAgent (Hong et al., 2023) can bridge the gap between frozen pre-trained language models and image encoders through trainable visual expert modules in the attention layer and FFN layer and can deeply fuse visual language features.

LLaVA-1.5 (Liu et al., 2023a) is an earlier proposed multi-modal large language model trained with multi-modal data generated by GPT-4.

GPT-4 Vision (OpenAI, 2023a) is the state-of-the-art multi-modal large language model with powerful complex reasoning and dialogue capabilities.

To compare the performance of the models more comprehensively, we use different strategies for comparison based on the above baseline model, such as CoT (Kojima et al., 2022) and self-consistency (Wang et al., 2022b).

4.1.3 Implementation Details

To obtain more accurate dialectical knowledge for subsequent knowledge transfer learning, we use the current optimal MLLM GPT-4 Vision to complete multi-agent debate. Its version is the Azure OpenAI 2023-12-01 preview version. Please refer to Appendix B.1 for more details.

Model	MMMU							MathVista					
	AD	Bus	Sci	HM	HS	TE	Total	FQA	GPS	MWP	TQA	VQA	Total
Qwen-VL	0.617	0.200	0.160	0.283	0.450	0.133	0.282	0.450	0.120	0.160	0.320	0.410	0.292
+ Zero CoT	0.550	0.133	0.133	0.360	0.467	0.105	0.264	0.420	0.090	0.170	0.380	0.360	0.284
+ Self Consistency	0.550	0.200	0.187	0.280	0.483	0.152	0.284	**0.470**	0.080	0.200	0.370	0.420	0.308
CogAgent	0.607	0.219	0.197	0.361	0.464	0.153	0.308	0.430	0.130	0.220	0.430	0.330	0.308
+ Zero CoT	0.571	0.192	0.211	0.403	0.500	0.143	0.310	0.420	0.120	0.240	0.420	0.310	0.302
+ Self Consistency	0.607	0.219	0.197	0.375	0.482	0.153	0.312	0.410	0.100	0.240	0.420	0.330	0.300
LLaVA-1.5 (7B)	0.483	0.147	0.227	0.373	0.467	0.200	0.298	0.130	0.070	0.130	0.300	0.280	0.182
+ Zero CoT	0.483	0.120	0.213	0.280	0.417	0.162	0.260	0.140	0.040	0.120	0.300	0.270	0.174
+ Self Consistency	0.467	0.160	0.200	0.333	0.500	0.181	0.286	0.150	0.030	0.120	0.320	0.290	0.182
LLaVA-1.5 (13B)	0.433	0.173	0.200	0.333	0.500	0.152	0.278	0.250	0.040	0.170	0.270	0.390	0.224
+ Zero CoT	0.467	0.173	0.133	0.293	0.533	0.162	0.271	0.150	0.060	0.130	0.320	0.420	0.216
+ Self Consistency	0.417	0.160	0.147	0.320	0.483	0.181	0.266	0.250	0.060	0.150	0.290	0.380	0.226
GPT4V	0.667	0.533	0.440	0.533	**0.783**	0.390	0.536	0.340	0.450	0.560	**0.620**	0.410	0.476
+ Zero CoT	0.617	**0.627**	0.400	0.560	0.767	0.381	0.538	0.460	0.410	**0.580**	0.590	0.420	0.492
+ Self Consistency	0.700	0.560	0.400	0.533	0.750	0.410	0.538	0.360	0.530	0.560	0.570	0.430	0.490
M2D (GPT4V)	**0.717**	0.587	**0.507**	**0.627**	0.767	**0.505**	**0.602**$_{\uparrow 6.4\%}$	0.460	**0.540**	**0.580**	0.580	**0.460**	**0.524**$_{\uparrow 3.2\%}$

Table 2: The main experiment result of the multi-modal multi-agent debate. The best results are highlighted in **bold**, while the second-best are underlined. The abbreviation in MMMU means AD: Art & Design, Bus: Business, Sci: Science, HM: Health & Medicine, HS: Humanities & Social Science, TE: Tech & Engineering. The abbreviation in MathVista means FQA: Figure QA, GPS: Geometry Problem Solving, MWP: Math Word Problem, TQA: Textbook QA, and VQA: Visual QA. Total represents the average of all items.

4.1.4 Main Results

The experimental results of our M2D and baseline methods are shown in Table 2. Based on the results, we have the following key findings:

(i) **Multi-modal multi-agent debates can significantly enhance the model's complex reasoning capabilities.** Experimental results demonstrate that our M2D method outperforms all baseline methods, achieving +6.4% and +3.2% improvement over the best-performing method on the MMMU and MathVista datasets, respectively. This demonstrates that debates can aid agents in reflecting on and rectifying errors during the reasoning process, uncovering logical relationships across different modalities, thereby improving the consistency and accuracy of the results, and generating high-quality dialectical knowledge.

(ii) **Open-source smaller MLLMs still lag significantly behind GPT-4 Vision**. Compared to GPT-4 Vision, current smaller MLLMs show an average performance gap of -25.5% on the MMMU dataset and -27.3% on the MathVista dataset. However, smaller MLLMs offer significant advantages in reasoning speed and deployment costs. Therefore, leveraging advanced MLLMs to enhance the capabilities of smaller MLLMs is crucial for accelerating their practical applications.

(iii) **Smaller MLLMs exhibit weak CoT performance**. Most smaller MLLMs exhibit varying degrees of decline after using CoT prompts. Our analysis identifies the main reasons as follows: (a) The multi-modal complex reasoning task requires comprehensive expert knowledge and multi-step complex reasoning for answers, but existing smaller MLLMs have weaker complex reasoning capabilities, often making mistakes in CoT reasoning's intermediate steps, which results in reasoning failures. (b) The lack of CoT-specific data in the fine-tuning datasets of current smaller MLLMs partially leads to inferior reasoning results. However, reasoning steps are essential for verifying the correctness and rationality of model inferences. Our method can condense high-quality dialectical knowledge and use it to fine-tune smaller models, significantly enhancing their logical reasoning and deep analysis capabilities. Detailed analysis is provided in section 4.2.5 and Appendix E.2.

4.1.5 Analysis of Debate Round and Agent Number

We sampled 100 data points each from the MMMU and MathVista datasets and analyzed the impact of debate rounds and agent numbers on the results to determine the most effective experimental setup.

Initially, we configured the system with two agents and varied the maximum debate rounds from 1 to 5, documenting the average number of rounds and their variance to assess the effect of round variability. Figure 3 displays that the average debate rounds increase before stabilizing at approximately 1.6, with statistical analysis showing that 98% of debates reach consensus within three rounds. Consequently, we maintained the maximum debate round

Model	Size	SFT Type	Data Type	AD	Bus	Sci	HM	HS	TE	Total
LLaVA-1.5	7B	LoRA	None	0.483	0.147	**0.227**	**0.373**	0.467	0.200	0.298
			Standard	**0.500**	0.213	0.160	0.347	0.433	0.200	$0.291_{\downarrow 0.7\%}$
			CoT	0.450	0.187	0.147	0.333	**0.533**	0.210	$0.291_{\downarrow 0.7\%}$
			CoD	0.450	**0.227**	0.173	0.320	0.500	**0.314**	$\mathbf{0.320}_{\uparrow \mathbf{2.2\%}}$
LLaVA-1.5	7B	Full	None	**0.483**	0.147	**0.227**	**0.373**	0.467	0.200	0.298
			Standard	0.367	0.200	0.173	0.333	0.500	**0.286**	$0.300_{\uparrow 0.2\%}$
			CoT	0.417	**0.293**	0.187	0.320	0.500	0.248	$0.313_{\uparrow 1.5\%}$
			CoD	0.417	**0.293**	**0.227**	0.293	**0.517**	**0.286**	$\mathbf{0.327}_{\uparrow \mathbf{2.9\%}}$
LLaVA-1.5	13B	LoRA	None	0.433	0.173	**0.200**	0.333	0.500	0.152	0.278
			Standard	0.483	0.213	0.173	0.253	0.550	0.152	$0.280_{\uparrow 0.2\%}$
			CoT	0.550	0.213	0.173	0.347	0.517	0.162	$0.302_{\uparrow 2.4\%}$
			CoD	**0.600**	**0.240**	0.187	**0.360**	**0.567**	**0.219**	$\mathbf{0.338}_{\uparrow \mathbf{6.0\%}}$
LLaVA-1.5	13B	Full	None	0.433	0.173	0.200	0.333	0.500	0.152	0.278
			Standard	0.417	0.240	0.240	0.240	0.550	0.181	$0.291_{\uparrow 1.3\%}$
			CoT	**0.467**	**0.267**	0.267	0.267	0.517	0.248	$0.322_{\uparrow 4.4\%}$
			CoD	**0.467**	0.213	**0.280**	**0.347**	**0.567**	**0.267**	$\mathbf{0.340}_{\uparrow \mathbf{6.2\%}}$

Table 3: The knowledge transfer learning results on the MMMU dataset. In data type, None denotes untrained original model, and Standard and CoT denote employing standard and CoT prompt data for knowledge transfer learning. The best results are highlighted in **bold**.

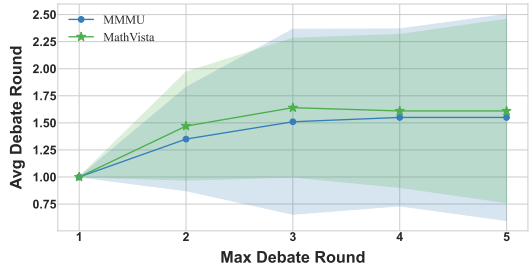


Figure 3: Average debate rounds under different maximum debate round settings on MMMU and MathVista datasets.

Agent Number	Max Round	MMMU	MathVista
1	3	0.54	0.49
2	3	0.59	**0.56**
3	3	0.62	0.54
4	3	0.62	0.52
5	3	**0.68**	0.51

Table 4: Performance as the agent number increases on MMMU and MathVista dataset.

at 3 for further experiments.

We examined the impact of varying agent numbers on debate outcomes across the MMMU and MathVista datasets, as detailed in Table 4. The results show that while the effectiveness of multi-agent debates on the MMMU dataset improves with more agents, it declines on the MathVista dataset. MMMU, a complex reasoning dataset spanning six core subjects, benefits from the diverse expertise brought by more agents, thereby enhancing model performance. In contrast, MathVista requires precise image understanding and multi-step reasoning, where inaccuracies introduced by large language models and increased by additional agents result in diminished performance, reflecting the observed downward trend.

4.2 Dialectical Knowledge Transfer Learning

4.2.1 Benchmark

To better evaluate our model, we used the same test set as the debate in Section 4.1.1. Moreover, to verify the model's multilingual generalization capability, we incorporated a Chinese multi-modal complex reasoning dataset, CMMMU (Zhang et al., 2024), similar to MMMU.

4.2.2 Baselines

To better evaluate the effectiveness of our method, we selected two models of different sizes: LLaVA-1.5 (Liu et al., 2023a) 7B and 13B, and validated them under two fine-tuning modes: LoRA (Hu et al., 2022) and full fine-tuning. Additionally, to better compare our transfer learning method based on chain-of-debate dialectical knowledge, we use GPT-4 Vision with standard and CoT prompts to generate reasoning processes and predicted results and employ them for knowledge transfer learning as our baseline.

4.2.3 Implementation Details

We randomly selected 3,000 pieces of data from the MMMU test set (without including the truth answers). and used GPT-4 Vision to generate debate data, with the number of agents and debate rounds set to 2 and 3. After filtering through No Answer, Wrong Answer, and Closest Answer, we obtained

2,275 pieces of data. Please refer to Appendix B.2 and B.3 for more details.

4.2.4 Main Results

We fine-tuned LLaVA-1.5 models of different parameter sizes using LoRA and full fine-tuning, with experimental results in Table 3. From the experimental results, it is observed that our models achieved the best results under different experimental setups. The LLaVA-1.5 7B and 13B models improved by +2.2% and +6.0%, respectively, under LoRA fine-tuning, and by +2.9% and +6.2% under full fine-tuning compared to the original models. The dialectical knowledge generated from multi-agent debates is of high quality and possesses a certain robustness, thereby validating the effectiveness of our method.

Further, we observed that under different experimental setups, the CoT knowledge transfer method outperformed the standard knowledge transfer method. This indicates that the logical reasoning process is more crucial for multi-modal complex reasoning tasks. Compared to the CoT knowledge transfer method, our approach achieved an average improvement of +2.4% across different settings, demonstrating that CoD condenses more logically robust dialectical knowledge. Transferring this knowledge to smaller models effectively boosted their logical reasoning abilities.

KTL Data Type	# CR	Total Score	Avg Score
Standard	131	331	2.52
CoT	131	353	2.69
CoD	**149**	**427**	**2.86**

Table 5: GPT-4 Vision scoring results for the inference process in the MMMU dataset. # CR represents the number of correct reasoning.

4.2.5 Analysis of CoD Reasoning Process

We initially selected case studies for analysis, with results indicating that our method enhances the model's accuracy and rationality in reasoning processes. Detailed analyses are available in Appendix F. Further, to assess the impact of dialectical knowledge, we fine-tuned the LLaVA-1.5 7B model using the LoRA method across three knowledge transfer datasets. We then employed GPT-4 Vision to evaluate the reasoning processes of various models on a scale from 1 to 5 for logic, depth, and detail, as detailed in Table 5. The results demonstrated that our CoD method, which incorporates dialectical knowledge, outperformed other approaches, suggesting that CoD effectively integrates and transfers dialectical knowledge into smaller MLLMs, enhancing their logical reasoning and deep analysis abilities.

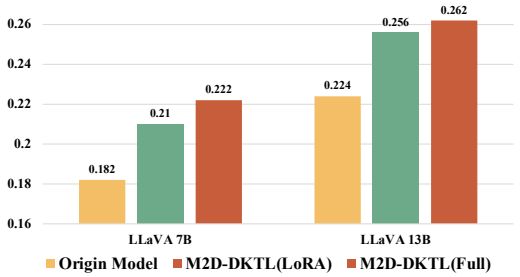


Figure 4: Results of generalization experiments conducted on the MathVista dataset.

4.2.6 Analysis of Generalization

To evaluate our model's generalization capabilities, we utilized LLaVA-1.5 fine-tuned with MMMU data in a CoD format and conducted tests on the MathVista dataset, with the experimental results depicted in Figure 4. The experimental results show that our model surpasses the origin MLLM across all configurations, with increases of +2.8%, +3.2%, +3.2%, and +2.2% on the LLaVA-1.5 7B LoRA, 7B full, 13B LoRA, and 13B full, respectively. The reason is that the MMMU dataset includes mathematics subject data, contributing to the model's enhanced generalization capabilities on the MathVista dataset for mathematical reasoning. Further experiments revealed that our method exhibits strong generalization capabilities in multilingual tasks. Detailed experimental results and analysis are provided in the Appendix E.1.

5 Conclusion

This study introduced a multi-modal multi-agent debate-based dialectical knowledge transfer learning framework (M2D-DKTL), which innovatively leverages multi-agent debates to enhance the knowledge quality and transfers them to bolster the reasoning capabilities of smaller MLLMs. The chain-of-debate method proved instrumental in condensing debate insights into a structured form that was effectively transferred to enhance the dialectical reasoning capacities of single-agent models. Systematic experimental results demonstrated that this method enhances the quality of knowledge, effectively refines dialectical knowledge, and transfers it to smaller models, significantly improving their logical reasoning capabilities and accelerating practical applications.

Limitations

Considering limitations in time and funding, we generated 3,000 data entries from the MMMU (Yue et al., 2023) test dataset for knowledge transfer learning. After filtering out noisy data, we obtained 2,275 data points, a relatively small dataset. However, compared to LoRA fine-tuning, full fine-tuning demands a higher quantity of data, which also resulted in our full fine-tuning experimental results not showing as significant an improvement as LoRA fine-tuning. Moreover, we verified our method's generalization on the MathVista (Lu et al., 2023) dataset for multi-modal mathematical complex reasoning and the CMMMU (Zhang et al., 2024) dataset for multilingual complex reasoning. To further promote the model's application, incorporating datasets from various domains for testing is necessary. Therefore, in future work, we plan to expand the training data volume for knowledge transfer learning to develop more effective smaller MLLMs. By teaching smaller MLLMs the dialectical knowledge found in multi-agent debates, we can boost their abilities in complex reasoning and hasten their real-world application.

References

Jinze Bai, Shuai Bai, Shusheng Yang, Shijie Wang, Sinan Tan, Peng Wang, Junyang Lin, Chang Zhou, and Jingren Zhou. 2023. Qwen-vl: A frontier large vision-language model with versatile abilities. *arXiv preprint arXiv:2308.12966*.

Chi-Min Chan, Weize Chen, Yusheng Su, Jianxuan Yu, Wei Xue, Shanghang Zhang, Jie Fu, and Zhiyuan Liu. 2024. Chateval: Towards better LLM-based evaluators through multi-agent debate. In *The Twelfth International Conference on Learning Representations*.

Yilun Du, Shuang Li, Antonio Torralba, Joshua B Tenenbaum, and Igor Mordatch. 2023. Improving factuality and reasoning in language models through multiagent debate. *arXiv preprint arXiv:2305.14325*.

Yao Fu, Hao Peng, Tushar Khot, and Mirella Lapata. 2023a. Improving language model negotiation with self-play and in-context learning from ai feedback. *arXiv preprint arXiv:2305.10142*.

Yao Fu, Hao Peng, Litu Ou, Ashish Sabharwal, and Tushar Khot. 2023b. Specializing smaller language models towards multi-step reasoning. In *Proceedings of the 40th International Conference on Machine Learning*, ICML'23. JMLR.org.

Zhi Gao, Yuntao Du, Xintong Zhang, Xiaojian Ma, Wenjuan Han, Song-Chun Zhu, and Qing Li. 2023. Clova: A closed-loop visual assistant with tool usage and update. *arXiv preprint arXiv:2312.10908*.

Muyang He, Yexin Liu, Boya Wu, Jianhao Yuan, Yueze Wang, Tiejun Huang, and Bo Zhao. 2024. Efficient multimodal learning from data-centric perspective. *arXiv preprint arXiv:2402.11530*.

Namgyu Ho, Laura Schmid, and Se-Young Yun. 2023. Large language models are reasoning teachers. In *Proceedings of the 61st Annual Meeting of the Association for Computational Linguistics (Volume 1: Long Papers)*, pages 14852–14882, Toronto, Canada. Association for Computational Linguistics.

Wenyi Hong, Weihan Wang, Qingsong Lv, Jiazheng Xu, Wenmeng Yu, Junhui Ji, Yan Wang, Zihan Wang, Yuxiao Dong, Ming Ding, et al. 2023. Cogagent: A visual language model for gui agents. *arXiv preprint arXiv:2312.08914*.

Cheng-Yu Hsieh, Chun-Liang Li, Chih-kuan Yeh, Hootan Nakhost, Yasuhisa Fujii, Alex Ratner, Ranjay Krishna, Chen-Yu Lee, and Tomas Pfister. 2023. Distilling step-by-step! outperforming larger language models with less training data and smaller model sizes. In *Findings of the Association for Computational Linguistics: ACL 2023*, pages 8003–8017.

Edward J Hu, yelong shen, Phillip Wallis, Zeyuan Allen-Zhu, Yuanzhi Li, Shean Wang, Lu Wang, and Weizhu Chen. 2022. LoRA: Low-rank adaptation of large language models. In *International Conference on Learning Representations*.

Lei Huang, Weijiang Yu, Weitao Ma, Weihong Zhong, Zhangyin Feng, Haotian Wang, Qianglong Chen, Weihua Peng, Xiaocheng Feng, Bing Qin, et al. 2023. A survey on hallucination in large language models: Principles, taxonomy, challenges, and open questions. *arXiv preprint arXiv:2311.05232*.

Takeshi Kojima, Shixiang Shane Gu, Machel Reid, Yutaka Matsuo, and Yusuke Iwasawa. 2022. Large language models are zero-shot reasoners. *Advances in neural information processing systems*, 35:22199–22213.

Junnan Li, Dongxu Li, Silvio Savarese, and Steven Hoi. 2023a. Blip-2: bootstrapping language-image pre-training with frozen image encoders and large language models. In *Proceedings of the 40th International Conference on Machine Learning*, ICML'23.

Liunian Harold Li, Jack Hessel, Youngjae Yu, Xiang Ren, Kai-Wei Chang, and Yejin Choi. 2023b. Symbolic chain-of-thought distillation: Small models can also "think" step-by-step. In *Proceedings of the 61st Annual Meeting of the Association for Computational Linguistics (Volume 1: Long Papers)*, pages 2665–2679.

Shiyang Li, Jianshu Chen, Yelong Shen, Zhiyu Chen, Xinlu Zhang, Zekun Li, Hong Wang, Jing Qian, Baolin Peng, Yi Mao, et al. 2022. Explanations from large language models make small reasoners better. *arXiv preprint arXiv:2210.06726*.

Tian Liang, Zhiwei He, Wenxiang Jiao, Xing Wang, Yan Wang, Rui Wang, Yujiu Yang, Zhaopeng Tu, and Shuming Shi. 2023. Encouraging divergent thinking in large language models through multi-agent debate. *arXiv preprint arXiv:2305.19118*.

Haotian Liu, Chunyuan Li, Qingyang Wu, and Yong Jae Lee. 2023a. Visual instruction tuning. In *Thirty-seventh Conference on Neural Information Processing Systems*.

Shilong Liu, Hao Cheng, Haotian Liu, Hao Zhang, Feng Li, Tianhe Ren, Xueyan Zou, Jianwei Yang, Hang Su, Jun Zhu, et al. 2023b. Llava-plus: Learning to use tools for creating multimodal agents. *arXiv preprint arXiv:2311.05437*.

Zhaoyang Liu, Zeqiang Lai, Zhangwei Gao, Erfei Cui, Zhiheng Li, Xizhou Zhu, Lewei Lu, Qifeng Chen, Yu Qiao, Jifeng Dai, et al. 2023c. Controlllm: Augment language models with tools by searching on graphs. *arXiv preprint arXiv:2310.17796*.

Pan Lu, Hritik Bansal, Tony Xia, Jiacheng Liu, Chunyuan Li, Hannaneh Hajishirzi, Hao Cheng, Kai-Wei Chang, Michel Galley, and Jianfeng Gao. 2023. Mathvista: Evaluating mathematical reasoning of foundation models in visual contexts. *arXiv preprint arXiv:2310.02255*.

OpenAI. 2023a. Gpt-4 technical report. *Preprint*, arXiv:2303.08774.

OpenAI. 2023b. Gpt-4v(ision) technical work and authors.

Xiao Pu, Mingqi Gao, and Xiaojun Wan. 2023. Summarization is (almost) dead. *arXiv preprint arXiv:2309.09558*.

Chen Qian, Xin Cong, Cheng Yang, Weize Chen, Yusheng Su, Juyuan Xu, Zhiyuan Liu, and Maosong Sun. 2023. Communicative agents for software development. *arXiv preprint arXiv:2307.07924*.

Noam Slonim, Yonatan Bilu, Carlos Alzate, Roy Bar-Haim, Ben Bogin, Francesca Bonin, Leshem Choshen, Edo Cohen-Karlik, Lena Dankin, Lilach Edelstein, et al. 2021. An autonomous debating system. *Nature*, 591(7850):379–384.

Quan Sun, Yufeng Cui, Xiaosong Zhang, Fan Zhang, Qiying Yu, Zhengxiong Luo, Yueze Wang, Yongming Rao, Jingjing Liu, Tiejun Huang, et al. 2023. Generative multimodal models are in-context learners. *arXiv preprint arXiv:2312.13286*.

Gemini Team, Rohan Anil, Sebastian Borgeaud, Yonghui Wu, Jean-Baptiste Alayrac, Jiahui Yu, Radu Soricut, Johan Schalkwyk, Andrew M Dai, Anja Hauth, et al. 2023. Gemini: a family of highly capable multimodal models. *arXiv preprint arXiv:2312.11805*.

Hugo Touvron, Louis Martin, Kevin Stone, Peter Albert, Amjad Almahairi, Yasmine Babaei, Nikolay Bashlykov, Soumya Batra, Prajjwal Bhargava, Shruti Bhosale, et al. 2023. Llama 2: Open foundation and fine-tuned chat models. *arXiv preprint arXiv:2307.09288*.

Haotian Wang, Xiyuan Du, Weijiang Yu, Qianglong Chen, Kun Zhu, Zheng Chu, Lian Yan, and Yi Guan. 2023a. Apollo's oracle: Retrieval-augmented reasoning in multi-agent debates. *arXiv preprint arXiv:2312.04854*.

Jianfeng Wang, Zhengyuan Yang, Xiaowei Hu, Linjie Li, Kevin Lin, Zhe Gan, Zicheng Liu, Ce Liu, and Lijuan Wang. 2022a. Git: A generative image-to-text transformer for vision and language. *arXiv preprint arXiv:2205.14100*.

Jiapeng Wang, Chengyu Wang, Xiaodan Wang, Jun Huang, and Lianwen Jin. 2023b. CocaCLIP: Exploring distillation of fully-connected knowledge interaction graph for lightweight text-image retrieval. In *Proceedings of the 61st Annual Meeting of the Association for Computational Linguistics (Volume 5: Industry Track)*, pages 71–80.

Peifeng Wang, Zhengyang Wang, Zheng Li, Yifan Gao, Bing Yin, and Xiang Ren. 2023c. SCOTT: Self-consistent chain-of-thought distillation. In *Proceedings of the 61st Annual Meeting of the Association for Computational Linguistics (Volume 1: Long Papers)*, pages 5546–5558.

Xuezhi Wang, Jason Wei, Dale Schuurmans, Quoc V Le, Ed H Chi, Sharan Narang, Aakanksha Chowdhery, and Denny Zhou. 2022b. Self-consistency improves chain of thought reasoning in language models. In *The Eleventh International Conference on Learning Representations*.

Jason Wei, Yi Tay, Rishi Bommasani, Colin Raffel, Barret Zoph, Sebastian Borgeaud, Dani Yogatama, Maarten Bosma, Denny Zhou, Donald Metzler, et al. 2022a. Emergent abilities of large language models. *Transactions on Machine Learning Research*.

Jason Wei, Xuezhi Wang, Dale Schuurmans, Maarten Bosma, Fei Xia, Ed H Chi, Quoc V Le, Denny Zhou, et al. 2022b. Chain-of-thought prompting elicits reasoning in large language models. In *Advances in Neural Information Processing Systems*.

Peter West, Chandra Bhagavatula, Jack Hessel, Jena Hwang, Liwei Jiang, Ronan Le Bras, Ximing Lu, Sean Welleck, and Yejin Choi. 2022. Symbolic knowledge distillation: from general language models to commonsense models. In *Proceedings of the 2022 Conference of the North American Chapter of the Association for Computational Linguistics: Human Language Technologies*, pages 4602–4625.

Kai Xiong, Xiao Ding, Yixin Cao, Ting Liu, and Bing Qin. 2023. Examining inter-consistency of large language models collaboration: An in-depth analysis via debate. In *Findings of the Association for Computational Linguistics: EMNLP 2023*, pages 7572–7590.

Xiang Yue, Yuansheng Ni, Kai Zhang, Tianyu Zheng, Ruoqi Liu, Ge Zhang, Samuel Stevens, Dongfu Jiang, Weiming Ren, Yuxuan Sun, et al. 2023. Mmmu: A massive multi-discipline multimodal understanding and reasoning benchmark for expert agi. *arXiv preprint arXiv:2311.16502*.

Ge Zhang, Xinrun Du, Bei Chen, Yiming Liang, Tongxu Luo, Tianyu Zheng, Kang Zhu, Yuyang Cheng, Chunpu Xu, Shuyue Guo, et al. 2024. Cmmmu: A chinese massive multi-discipline multimodal understanding benchmark. *arXiv preprint arXiv:2401.11944*.

Lianmin Zheng, Wei-Lin Chiang, Ying Sheng, Siyuan Zhuang, Zhanghao Wu, Yonghao Zhuang, Zi Lin, Zhuohan Li, Dacheng Li, Eric Xing, et al. 2023. Judging llm-as-a-judge with mt-bench and chatbot arena. *arXiv preprint arXiv:2306.05685*.

A Role Definition

During the process of multi-modal multi-agent debates, there are three primary roles: the debater, the judge, and the summarizer. Each performs their respective duties, collaboratively completing the debate and achieving a consistent and correct reasoning result. In the debate process, we distinguish the roles of debaters in different rounds, with the definitions of different roles as follows.

The definition of debater in express stance phase is:

> *You're an intelligent, diplomatic, and assertive debater. Your current responsibility is to generate your own initial views for use in the subsequent debate process.*

The definition of debater in orderly debate phase is:

> *You're an intelligent, diplomatic, and assertive debater. Your responsibility is to debate with other agents to collaboratively determine the correct answers to various questions. Uphold your position when confident of its accuracy. If you detect a mistake in your reasoning, promptly admit and correct it with accurate information.*

The definition of judge role is:

> *As a judge agent, your primary responsibility if to impartially evaluate the response of other agents for consistency. Please ensure your judgements are objective, replying solely on the coherence and alignment of the provided answers.*

The definition of summarizer role is:

> *You are an intelligent agent, tasked with synthesizing the response of other agents into a concise and comprehensive final answer. Your role is not to generate original responses, but to condense the information provided by other agents into a succinct summary.*

B Implementation Details

B.1 Multi-Modal Multi-Agent Debate

Our multi-modal multi-agent debate experiment was conducted using GPT-4 Vision, provided by Azure OpenAI 12-01-preview version. Additionally, we employed open-source multi-modal language models Qwen-VL[2] (Bai et al., 2023), CogAgent[3] (Hong et al., 2023), LLaVA-1.5 7B[4] (Liu et al., 2023b), LLaVA-1.5 13B[5] (Liu et al., 2023b), and GPT-4 Vision (OpenAI, 2023a) as our baseline models using Standard Prompt B.2. Furthermore, to thoroughly compare our model, we conducted experimental comparisons based on each model using CoT Prompt B.3 and the Self-Consistency strategy. Specific hyperparameters are listed in Table 6. In addition, we use the Azure OpenAI 12-01-preview version of GPT-4 to complete the answer evaluation.

Hyperparameters	Values
# Agent	2
# Deabte Round	3
# max tokens	1024
Self-consistenty	3
Temperature	1.0

Table 6: Hyperparameters of multi-modal multi-agent debate.

Filter Rule	# Examples
No Answer	191
Wrong Answer	34
Closest Answer	320
Multi Images	179

Table 7: The number filtered out by different filtering rules.

B.2 Dialectical Data Construction

To enhance the accuracy of dialectical data, we first filter the debate data using three different filtering rules, with the specific results shown in Table 7. To better enable smaller MLLMs to learn the dialectical knowledge produced by debates inspired by CoT (Wei et al., 2022b), we input the content of multi-agent debates, along with questions and images, into GPT-4 Vision (OpenAI, 2023a). This allows it to generate reasoning chains containing dialectical knowledge for knowledge transfer learn-

[2] https://huggingface.co/Qwen/Qwen-VL-Chat
[3] https://huggingface.co/THUDM/cogagent-vqa-hf
[4] https://huggingface.co/liuhaotian/llava-v1.5-7b
[5] https://huggingface.co/liuhaotian/llava-v1.5-13b

Hyperparameters	7B		13B	
	LoRA	Full	LoRA	Full
Batch Size (Per Device)	8	2	8	1
Gradient Accumulation Steps	1	4	1	8
Learning Rate	5e-5	2e-5	5e-5	2e-5
Max Length	2048	2048	2048	2048
# GPUs	2	4	2	4

Table 8: Hyperparameters of knowledge transfer learning.

ing, which we call the chain-of-debate. Specific prompt in B.1.

Prompt B.1: Construct Dialectical Data

Based on the question and the answers of all agents, extract the reasoning process in the form of a logical and concise chain of thought. Put the final answer in the square brackets.
HERE IS AN EXAMPLE OF ANSWER:
{**an_example**}
(END OF EXAMPLE)

QUESTION: {**question**}
DEBATE PROCESS:
{**debate_process**}

ANSWER:

B.3 Dialectical Knowledge Transfer Learning

In the process of knowledge transfer learning, we chose LLaVA-1.5 (Liu et al., 2023b) as our base model and conducted fine-tuning in both LoRA and Full modes on two-parameter scales, 7B and 13B, respectively. During the model fine-tuning and inference phases, we used A800 80G GPUs. In the inference process, we set the temperature and max tokens to 0.2 and 1024, respectively. The specific hyperparameter settings are shown in Table 8. To better compare the effect of our Chain-of-Debate integrated with dialectical knowledge, we also used Standard Prompt B.2 and CoT Prompt B.3 to obtain inference results based on GPT-4 Vision, filtered them using the same filtering method as CoD and used them to fine-tune the LLaVA model.

Prompt B.2: Construct Standard Data

Please fully understand the information in the image and answer the following question accurately as possible. Put the final answer in the square brackets at the end of your response.

QUESTION: {**question**}
ANSWER:

Prompt B.3: Construct Chain-of-Thought Data

Please fully understand the information in the image and answer the following question step by step as accurately as possible. Put the final answer in the square brackets at the end of your response.

QUESTION: {**question**}
ANSWER:

Table 9: Statistics for each question type in MMMU, MathVista, and CMMMU datasets.

Dataset	# Total Examples	SubClass	# Examples
MMMU	450	Art & Design	60
		Business	75
		Science	75
		Health & Medicine	75
		Humanities & Social Science	60
		Tech & Engineering	105
MathVista	500	Figure QA	100
		Geometry Problem Solving	100
		Math Word Problem	100
		Textbook QA	100
		Visual QA	100
CMMMU	448	Art & Design	44
		Business	63
		Science	102
		Health & Medicine	75
		Humanities & Social Science	42
		Tech & Engineering	122

C Details of Datasets

We introduce the three datasets used in the paper and provide a detailed statistical result, as shown in Table 9.

C.1 MMMU Dataset

MMMU (Yue et al., 2023) is a multi-modal QA dataset collected from university exams, quizzes, and textbooks. It covers six core subjects: Art & Design, Business, Science, Health & Medicine, Humanities & Social Science, and Tech & Engineering. See Figure 5 for data examples. MMMU focuses on advanced perception and reasoning with domain-specific knowledge, challenging models to perform tasks akin to those experts face. The dataset is divided into dev, validation, and test sets. The dev dataset contains 150 entries for few-shot settings, the test dataset consists of 10,500 entries without groundtruth, and the validation set includes 900 entries with groundtruth. Consequently, we randomly selected 50% of the data from each category in the validation set, resulting in 450 entries for evaluating our model.

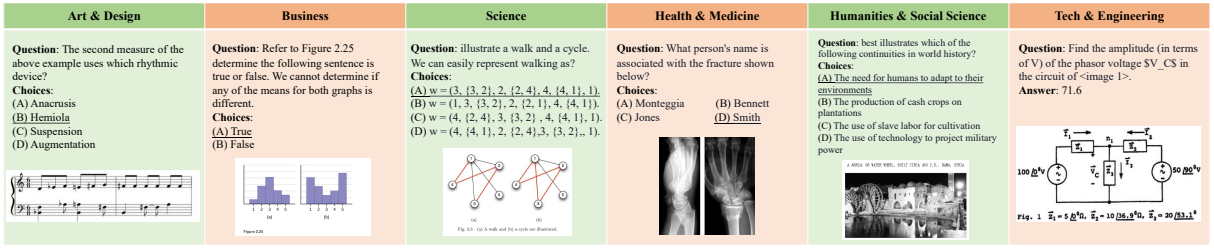


Figure 5: Examples of MMMU dataset.

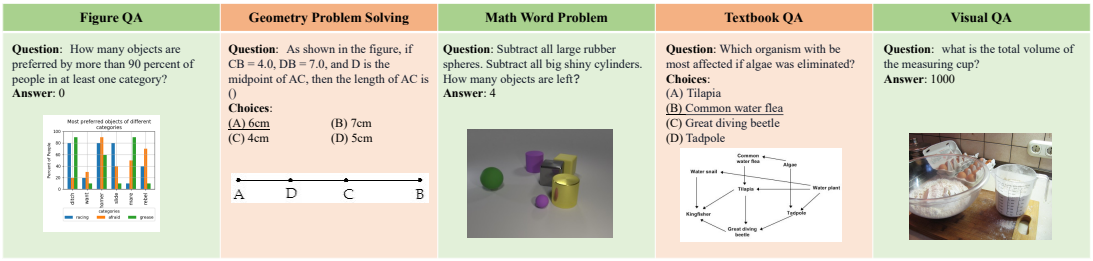


Figure 6: Examples of MathVista dataset.

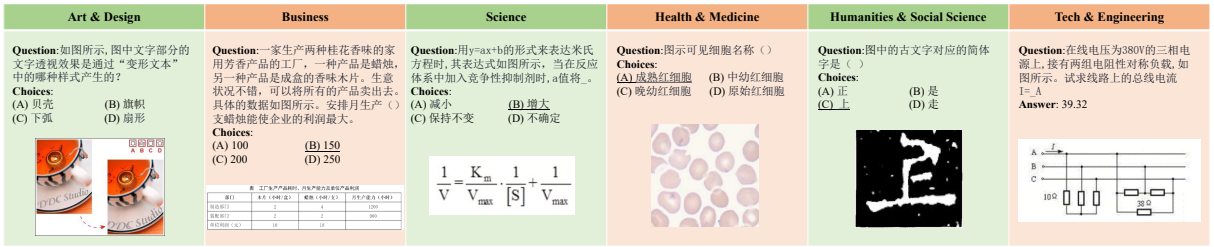


Figure 7: Examples of CMMMU dataset.

C.2 MathVista Dataset

MathVista (Lu et al., 2023) is a multi-modal complex reasoning dataset in the mathematics domain, featuring six types of tasks: figure QA, geometry problem solving, math word problem, textbook QA, and visual QA. It requires fine-grained, in-depth visual understanding and combinational reasoning designed to assess the capabilities of visual models in mathematical reasoning tasks. See Figure 6 for data examples. The dataset is divided into test and testmini subsets, where the test dataset contains 5141 entries without groundtruth for online evaluation, and the testmini contains 1000 entries with groundtruth, totaling 6141 entries. Therefore, we randomly selected 50% of the data from each category in the testmini dataset for model evaluation.

C.3 CMMMU Dataset

Similar to MMMU, CMMMU (Zhang et al., 2024) is a Chinese multi-modal complex reasoning benchmark designed to assess the performance of visual models on tasks requiring university-level subject knowledge and complex reasoning in a Chinese context. See Figure 7 for data examples. The dataset contains 12k entries, divided into dev, validation, and test datasets. The dev dataset includes 110 entries for few-shot settings, the test dataset consists of 11,000 entries without groundtruth, and the validation set contains 900 entries with groundtruth. Therefore, we randomly extracted about 50% of the data from each category in the Validation set, resulting in 448 entries for evaluating our model.

Model	MMMU		MathVista	
	# Tokens	Acc	# Tokens	Acc
Standard	901	0.536	673	0.476
CoT	1,026	0.538	751	0.492
Self-consistency	2,708	0.538	2,022	0.490
Our Method	4,439	0.602	3,319	0.524
Upper Limit	~8,000	–	~6,000	–

Table 10: Average token consumption of different models.

D Token Cost

We conducted a statistical analysis of the token expenditure by GPT-4 Vision under different settings, with the results presented in Table 10. Compared to the Standard Prompt, we observed that the Chain of Thought (CoT) consumes more output tokens, indicating that CoT can generate a more compre-

hensive chain of reasoning. The experimental findings reveal that CoT yields a greater improvement in MathVista than in MMMU, attributable to MathVista's higher demands on the reasoning process. Our approach sets the number of agents to 2 and the maximum debate rounds to 3. Our statistical analysis indicates that although our method incurs a slight increase in token expenditure compared to the self-consistency approach, the performance enhancement is significantly more pronounced. **We believe that the performance gains justify this increase in token cost.** Additionally, we estimated the token expenditure when using the maximum number of debate rounds as the termination condition and found our method significantly reduces token costs compared to this approach, making it a more efficient alternative.

E Details of Analysis

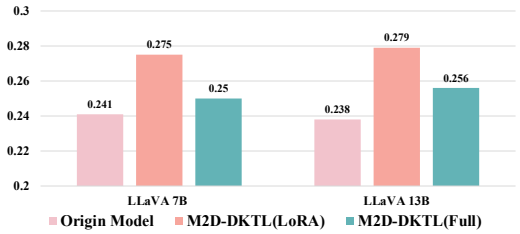


Figure 8: Results of multilingual generalization experiments conducted on the CMMMU dataset.

E.1 Analysis of Multilingual Generalization

Multilingual capability is an essential aspect in evaluating a model's generalization ability. To analyze our model's generalization in the multilingual domain, we selected a Chinese multi-modal complex reasoning dataset (CMMMU), aiming to assess interdisciplinary understanding and reasoning abilities in the Chinese context. As shown in Figure 8, we tested the LLaVA-1.5 model under different experimental settings. The results demonstrate that our model outperforms the baseline across all settings, **proving that our model still possesses strong generalizability in multilingual settings.** Additionally, the results indicate that LoRA fine-tuning outperforms full fine-tuning on the CMMMU dataset, suggesting that the gains from full fine-tuning are less significant than those from LoRA fine-tuning. The main reason is that the current knowledge transfer datasets are limited, and larger datasets are advantageous for full fine-tuning. However, full fine-tuning still surpasses the performance of the original model. To further improve the model's generalization across different fields, we can use our model to construct datasets from various domains and fine-tune the model, thereby enhancing its domain generalization capabilities.

Model	Easy	Medium	Hard
Qwen-VL	0.388	0.225	0.141
CogAgent	0.415	0.304	0.138
LLaVA-1.5 (7B)	0.388	0.268	0.153
LLaVA-1.5 (13B)	0.388	0.244	0.106
GPT4V	0.684	0.540	0.282
M2D (GPT4V)	**0.743**	**0.549**	**0.482**

Table 11: Result decomposition of multi-modal multi-agent debate across question difficulty levels on the MMMU dataset.

Model	Size	SFT Type	Data Type	Easy	Medium	Hard
LLaVA-1.5	7B	LoRA	None	0.421	0.281	0.141
			Standard	0.408	0.239	0.212
			CoT	0.342	**0.291**	0.200
			CoD	**0.428**	0.254	**0.294**
LLaVA-1.5	7B	Full	None	**0.421**	0.281	0.141
			Standard	0.407	0.253	0.223
			CoT	0.408	0.291	0.200
			CoD	0.382	**0.329**	**0.224**
LLaVA-1.5	13B	LoRA	None	0.421	0.244	0.105
			Standard	0.362	0.268	0.164
			CoT	0.394	0.291	0.165
			CoD	**0.454**	**0.300**	**0.226**
LLaVA-1.5	13B	Full	None	0.421	0.244	0.105
			Standard	0.368	0.300	0.129
			CoT	**0.441**	0.291	0.188
			CoD	0.414	**0.323**	**0.247**

Table 12: Result decomposition across question difficulty levels on the MMMU dataset. The best results are highlighted in **bold**.

E.2 Dialectical Knowledge is More Beneficial for Complex Problems

Tables 11 and 12 compare the performance of our multi-modal multi-agent debate framework (M2D) and dialectical knowledge transfer method against baseline methods across three difficulty levels in the MMMU dataset. Table 11 shows that our proposed method (M2D) excels in handling complex multi-modal tasks, particularly at higher difficulty levels, achieving significantly higher accuracy than baseline methods. This advantage stems from the multi-agent debate process, which **effectively uncovers and integrates deep logical relationships across modalities**, enhancing the model's accuracy and understanding depth in complex problem-solving. Additionally, Table 12 demonstrates that the dialectical knowledge transfer method outper-

Model	Prompt Type	# Output Tokens	# ESR
LLaVA-1.5 7B	Standard	74	53
LLaVA-1.5 7B	CoT	130	45
LLaVA-1.5 13B	Standard	85	58
LLaVA-1.5 13B	CoT	146	38
GPT4 Vision	Standard	242	0
GPT4 Vision	CoT	365	0

Table 13: Average output tokens and extremely short responses (ESR) number of LLava-1.5 and GPT4 Vision.

forms baseline methods, especially for more difficult problems. By using the CoD strategy, the model refines and transfers dialectical knowledge from multiple debate rounds, **effectively reducing information distortion and error accumulation**, thereby improving the logical reasoning and decision-making capabilities of smaller models in multi-modal environments. These findings confirm the effectiveness of the chain-of-debate technique in enhancing model performance on complex multi-modal tasks, particularly for high-difficulty problems, showcasing the significant advantages and practical potential of our approach.

E.3 LLaVA 7B Performs Better Than 13B in MMMU

From the experimental results of LLaVA-1.5 7B and 13B on MMMU and CMMMU, we found that the performance of LLaVA-1.5 7B is superior to the larger-parameter LLaVA-1.5 13B. After analyzing the datasets and experimental results, we discovered that MMMU and CMMMU are multi-modal reasoning datasets requiring multifaceted expert knowledge. LLaVA-1.5 13B has a clearer self-awareness and will refuse to answer when it perceives its knowledge as insufficient to reason the correct results, as seen in instances like "*However, the exact cause of the decline cannot be determined solely from the image.*" and "*The p-value is not directly visible in the image, but it is likely to be related to the data presented in the table.*". Conversely, LLaVA-1.5 7B tends to provide short answers without a reasoning process, enabling it to correctly guess some answers, thus slightly outperforming LLaVA-1.5 13B.

E.4 Smaller MLLMs are not Suitable for Debate

We have compiled statistics on the average output tokens and the quantity of extremely short responses for LLaVA-1.5 7B, LLaVA-1.5 13B, and GPT-4 Vision, with the specific data presented in the Table 13. Compared to GPT-4 Vision, LLaVA models produce shorter responses, which disadvantage agents in completing debates. Additionally, we randomly selected 100 data from the reasoning results of each model to count the number of extremely short responses. We found that, when using the same prompt, LLaVA-1.5 7B and 13B are more prone to generating extremely short responses, often containing just an option or a phrase, a situation not encountered with GPT-4 Vision. Such extremely short responses, lacking a reasoning process, make the debate meaningless. This issue arises because LLaVA-1.5 has smaller parameter sizes, poorer instruction-following capabilities, and is not specifically fine-tuned for multi-turn dialogue tasks, leading the model to prefer shorter responses.

F Case Study

We compared our model fine-tuned with LLaVA 13B LoRA to an unfine-tuned model and displayed more examples under different datasets, as shown in Figure 9. The results reveal that the original LLaVA 13B model's generated answers can exhibit issues such as a lack of a reasoning process or errors in reasoning. After fine-tuning with our constructed dialectical data, the model can generate accurate reasoning processes and answers, providing reliable support for the model's interpretability. This demonstrates that **our method can endow smaller models with strong logical reasoning and deep analysis capabilities**.

G Prompts

G.1 Debate Process Prompts

In the multi-modal multi-agent debate process, there are essentially two phases. Initially, debaters express their stances (Prompt in G.1), and then a judge determines whether these positions are consistent (Prompt in G.3). If not, the debate enters an orderly debate phase (Prompt in G.2). After each round of orderly debate, the judge assesses whether the debaters have reached consistency. The debate ends when a consensus or the maximum number of rounds is reached. Finally, the summarizer compiles the viewpoints of the debaters to form the ultimate conclusion(Prompt in G.4).

G.2 GPT4 Evaluation Prompt

GPT-4 (OpenAI, 2023a) has demonstrated remarkable capabilities, with Zheng et al. (2023) exploring

Prompt G.1: Debater (Express Stance)

SYSTEM: {**first_round_debate_role_definition**}

INSTRUCTION: Thoroughly analyze the information presented in the image. Answer the question step by step as accurately as possible. Put the final answer in the square brackets at the end of your response.

QUESTION: {**question**}
ANSWER:

Prompt G.2: Debater (Orderly Debate)

SYSTEM: {**other_round_debate_role_definition**}

INSTRUCTION: Reflect your historical response by analyzing the image in detail. Then engage in a step by step debate with other agents, considering their viewpoints. Extract the essence and discard the dregs. Work with other agents to improve each other's reasoning process. Each response should build on the previous ones, focusing on accuracy and a logical progression of ideas based on the image's information. Put the final answer in the square brackets at the end of your response.

{**answer_from_other_agents**}

{**your_historical_answer**}

QUESTION: {**question**}
ANSWER:

Prompt G.3: Judge

SYSTEM: {**judge_role_definition**}

INSTRUCTION: Review the agents' answers and judge their consistency. Respond with [Yes] if the answers are consistent, or [No] if they are not. Make sure to enclose your response of Yes or No within square brackets.

HERE IS AN EXAMPLE:
{**an_example**}
(END OF EXAMPLES)

QUESTION: {**question**}
{**all_answers_from_agents**}
ANSWER:

Prompt G.4: Summarizer

SYSTEM: {**summarizer_role_definition**}

INSTRUCTION: Please summarize the final answer from answer of all agents. Generate your response step by step and place the final answer in the square brackets at the end of response.

HERE IS AN EXAMPLE:
{**an_example**}
(END OF EXAMPLES)

QUESTION: {**question**}
{**all_answers_from_agents**}
SUMMARY:

its performance in evaluation tasks, proving the potential of GPT-4 for generating content evaluations. The multi-modal complex reasoning dataset used in this work includes multiple-choice and open-ended questions with relatively short answers. Given reference answers, GPT-4 can effectively undertake the task of evaluating answers with the GPT-4 evaluation prompt provided in G.5.

G.3 GPT-4 Vision Evaluation Prompt

To evaluate the logic and rationality of the model's reasoning process, we referred to the evaluation work (Zheng et al., 2023) and used GPT-4 Vision to score the reasoning processes of models after different knowledge transfer learning. The scoring was based on three aspects: logic, depth, and detail levels, with a scoring range from 1 to 5. The specific prompt is as described in Prompt G.6.

Prompt G.5: GPT-4 Eval

You are a judge agent, your primary responsibility is to impartially evaluate the evaluation answer for correct. Please judge the correctness of the llm answer based on the reference answer. Return [True] if the llm answer matches one of the reference answer, otherwise return [False]. Make sure that True and False are placed in brackets.

QUESTION: {**question**}
REFERENCE ANSWER: {**reference_answer**}
LLM ANSWER: {**llm_answer**}
JUDGE ANSWER:

Prompt G.6: GPT-4 Vision Score

Please act as an impartial judge and evaluate the quality of the response provided by an AI assistant to the user question displayed below. Your evaluation should consider the logical, depth, and level of detail of the reasoning process. Be as objective as possible. Please rate the reasoning process on a scale of 1 to 5 by strictly following this format: "Rating: [score]", for example: "Rating: [3]".

QUESTION: {**question**}
REFERENCE ANSWER: {**reference_answer**}
LLM REASONING RESULT: {**llm_answer**}

(MMMU) validation_Pharmacy_9



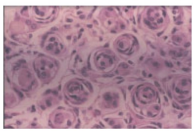
Question:
Look at the two pictures below. Which picture appears to have more order?<image 1>
Choices: (A) a (B) b (C) same of the two
GroundTruth: (A) a

LLaVA-1.5 13B: a

M2D-DTKL(LLaVA-1.5 13B LoRA SFT):
LOGICAL CHAIN OF DEBATE:
1. Order is defined as the degree of structure, organization, or arrangement in a system or environment.
2. Image 1 shows a jar of marbles organized in a pattern within a container, indicating some level of order.
3. Image 2 shows marbles scattered disorganized on the floor, lacking any visible structure or arrangement, indicating a lower level of order.
4. By comparison, Image 1 appears to have more order than Image 2 due to the marbles being contained and organized in a jar.

Final Answer: [**(A) a**]

(MMMU) validation_Design_7



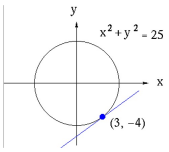
Question:
This tumor is: <image 1>
Choices: (A) more common in children than adults (B) may be intraventricular in children
(C) is often associated with seizures (D) also has binucleate neurons
GroundTruth: (B) may be intraventricular in children

LLaVA-1.5 13B: This image shows a close-up view of a tumor, which is more common in children than adults. The tumor is purple in color and has a cellular structure. It is not clear from the image whether it is intraventricular in children or if it is associated with seizures. However, the image does show that **the tumor has binucleate neurons**.

M2D-DTKL(LLaVA-1.5 13B LoRA SFT):
LOGICAL CHAIN OF DEBATE:
1. The image provided is a histological slide of a brain tumor, but it does not include clinical information about the patient's age, seizure history, or tumor location.
2. Without additional clinical data, it is not possible to determine the age group commonly affected by this tumor type (A).
3. The image does not show binucleate neurons, which are characterized by having two nuclei, so option (D) is incorrect.
4. The presence of rosenthal fibers suggests an intraventricular tumor, which is common in children (B).
5. Seizures are a common clinical presentation of brain tumors, but the image alone does not confirm the association with seizures (C).
6. The most accurate answer, based on the histological features observed, is that the tumor may be more common in children than adults and may be intraventricular in children.

Final Answer: [**(B) may be intraventricular in children**]

(MathVista) 767



Question:
What is the maximum value of y?

GroundTruth: 5

LLaVA-1.5 13B: The maximum value of y can be determined by looking at the equation and the coordinates of the points on the graph. The equation is y = 3x^2 - 4x + 3. To find the maximum value of y, we need to find the vertex of the parabola. The x-coordinate of the vertex can be found by solving the equation x = -b/2a, where a = 3 and b = -4. Plugging these values into the equation, we get x = -(-4)/(2*3) = 2/6 = 1/3. The y-coordinate of the vertex can be found by plugging this value into the equation y = 3x^2 - 4x + 3. So, **the maximum value of y is 3(1/3)^2 - 4(1/3) + 3 = 1/9. [1/9]**

M2D-DTKL(LLaVA-1.5 13B LoRA SFT):
LOGICAL CHAIN OF DEBATE:
1. The image shows a graph with a circle and a right-angled triangle, and the equation x^2 + y^2 = 25 is given.
2. The radius of the circle is 5, and the area of the circle is A = πr^2, so A = π(5)^2 = 25.
3. The equation represents a circle, and any point on the circle is equally distant from the center, so y = ±5.
4. The maximum value of y is reached when y is most positive, which is when y = 5.
5. The maximum value of y is 5.

Final Answer: [**5**]

Figure 9: Case study in MMMU and MathVista datasets.